Are Digital Spinners Fair?

The Median Market is a new grocery store in town with a fun checkout process. After every purchase, customers spin a digital spinner for a chance to win a discount or a free item. Mrs. Gallas is skeptical of digital spinners and decides to keep track of her results. Over her next 24 spins at checkout, she wins a prize (discount or free item) 17 times. $17 < 20$

$np = 24 \times \frac{5}{6}$

Some evidence that spinner is unfair



1. How many times would you expect the spinner to land on a prize in 24 spins? Is it guaranteed?

20 prizes. Not guaranteed to get exactly 20 prizes.

Do the results from Mrs. Gallas' 24 spins give convincing statistical evidence that the digital spinner is not fair? We will perform a **simulation** to help answer this question. We will **assume the spinner is fair**. If the spinner is fair, what is the probability of winning a prize in 24 spins **purely by chance?** Let's find out.

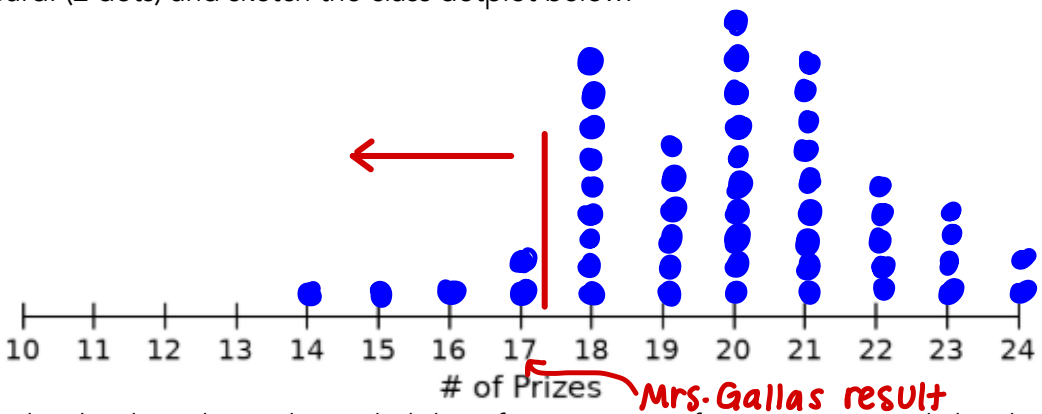
2. What could we use to model the probability? roll a die Assign certain outcomes to winning a prize or "Maybe next time!". List them below.

1 → "maybe next time!" 2,3,4,5,6 → prizes

3. Roll your die 24 times to imitate the process of Mrs. Gallas spinning the digital spinner. How many of them were a prize? 15

4. Do another simulation of 24 spins. How many were a prize this time? 16 } Answers vary

5. Plot the number of times a prize is won for each simulation of 24 spins to the dotplot on the board. (2 dots) and sketch the class dotplot below.



6. Based on the dotplot, what is the probability of getting 17 or fewer prizes, purely by chance?

5/50 = 0.10 or 10%

7. Does it seem plausible that the digital spinner is fair, but that Mrs. Gallas just got unlucky? Or do we have convincing statistical evidence that the Median Market is using an unfair spinner? Explain.

It is plausible that the spinner is fair and Mrs. G's result happened purely by chance.

10% > 5% → not statistically significant

Lesson 3.2 – Simulation

QuickNotes

LT#1 Simulation

A way to model random events, such that simulated outcomes closely match real world outcomes.

Ex: roll die, toss coin, applet, RNG

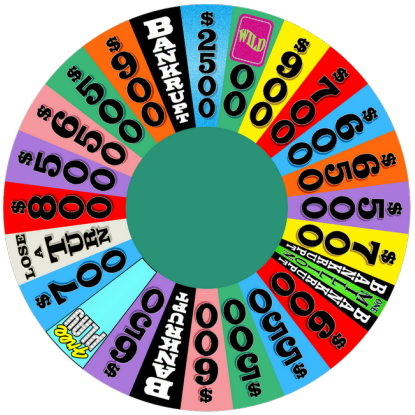
LT#2 Evidence for a claim

Assuming a claim is true, find the probability of getting the observed result or more extreme.

< 5% → • statistically significant
• convincing evidence against the claim

Check Your Understanding

Some people have wondered if the Wheel of Fortune wheel is fair. If it is fair, the "Bankrupt" result should occur with probability 1/9. To test the wheel, Pat Sajak observed a random sample of 50 spins of the wheel and found that "Bankrupt" was spun 10 times. Is this convincing evidence that Bankrupt comes up more often than it should?



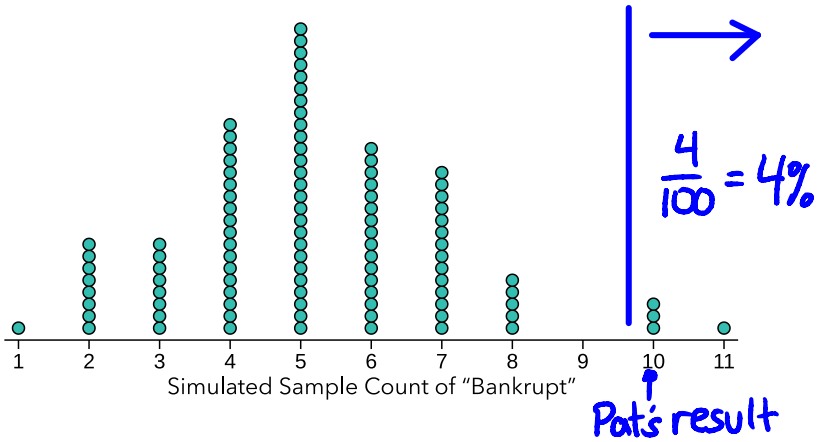
a. If the wheel is fair, how many "Bankrupt" would you expect in 50 spins?

$50 \times 1/9 = 5.56$ Bankrupts

b. Does Pat Sajak have some evidence that the wheel is unfair? Why?

Yes because $10 > 5.56$

The dotplot displays the results of 100 simulated trials of 50 spins of a fair wheel.



c. Explain what the one dot above 11 indicates.

In one simulation of 50 spins of a fair wheel, there were 11 Bankrupts.

d. Assuming the wheel is fair, what is the estimated probability of getting 10 or more bankrupt in 50 spins?

$4/100 = 0.04$ or 4%

e. Does Pat Sajak have convincing evidence that the wheel is unfair? Explain your answer.

Yes. Assuming a fair wheel, there is only a 4% probability of getting 10 or more Bankrupts purely by chance.

Because this is unlikely, Pat has convincing evidence of an unfair wheel.